

Homework 1

Submission date: 1st April, 2019

6. The stress tensor of a point is given as $\begin{bmatrix} 12 & 6 & 0 \\ 6 & 10 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, Calculate the maximum principle stress and its direction.

7. Do the following two stress tensors represent the same state of stress? Why?

$$\sigma_{ij}^1 = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \sigma_{ij}^2 = \begin{bmatrix} \frac{a+b}{2} & \frac{a-b}{2} & 0 \\ \frac{a-b}{2} & \frac{a+b}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

8. (For isotropic elastic body) Calculate **the difference** between normal stresses **and the difference** between normal strains. Then (1) Derive the relationship between **them**. (2) Prove that when principle stress $\sigma_1 \geq \sigma_2 \geq \sigma_3$, then its principal strain $\varepsilon_1 \geq \varepsilon_2 \geq \varepsilon_3$.

Tips:

The generalized Hooke's law under isotropic condition is

$$\varepsilon_{xx} = \frac{1}{E} [\sigma_{xx} - \nu(\sigma_{yy} + \sigma_{zz})]$$

$$\varepsilon_{yy} = \frac{1}{E} [\sigma_{yy} - \nu(\sigma_{xx} + \sigma_{zz})]$$

$$\varepsilon_{zz} = \frac{1}{E} [\sigma_{zz} - \nu(\sigma_{xx} + \sigma_{yy})]$$